CODERED ’26 Challenge Brief: Customer Knowledge Graph for Hybrid B2B Commerce

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Challenge at a glance

Build a centralised customer knowledge graph that lets a hybrid online-plus-field-sales B2B company in India know who its customers are, what they are working on, and whom to call next.

Challenge Partner	An Indian B2B e-commerce marketplace for robotics, industrial automation, electronic components, drones and 3D printing, selling online and through a field sales team. The company is not named in this brief.
Domain	Knowledge graphs, entity resolution, data engineering, applied AI, B2B sales enablement
Users of the product	Field sales, inside sales, marketing, leadership
Data	Open and public sources. Teams simulate the company’s internal data.
Format	About one month of remote build, two online evaluation rounds, then a 24-hour on-campus sprint at BMSITM on 3 and 4 December 2026
What we want to see	A working graph and the four sales workflows it powers (account brief, territory query, lead discovery, next-need signals)

About the Challenge Partner

The partner is an India-based B2B marketplace with a large multi-brand catalogue spanning robotics, industrial automation, electronic components, drones and 3D printing. Its customers include manufacturers, system integrators, defence suppliers, universities, research labs, startups and individual engineers.

Sales happen two ways. Most orders arrive online through the website, where a buyer creates an account, browses, and checks out with a GST number and a shipping address. A field sales team handles the larger and repeat business: it visits customers, scopes projects and quotes against requirements.

Every order, web session and sales visit leaves a trail of data. Today that trail sits in separate systems and in salespeople’s heads. The partner wants it connected.

Why this is hard in India

A B2B buyer in India is known only in fragments: a GSTIN on an invoice, a PIN code on a shipping label, a personal Gmail address on a web account, and a company name spelled three different ways across three orders. Nothing ties those fragments together.

In the US or Europe a company would buy that stitching from a data aggregator and get a ready company profile with industry, size, location and contacts. India has no equivalent with real depth on SMEs, integrators, labs and startups. The information exists, but it is scattered across the GST portal, MCA company filings, Udyam registrations, customs and import records, tender portals, company websites, job posts and news, each in its own format and with its own quirks.

The internal side is just as fragmented. The order system knows what was bought and where it was shipped. Web analytics knows what was browsed, from which device and IP. The CRM, where one exists, knows what a salesperson typed after a meeting. None of these share an identifier.

So a sales rep walks into a meeting knowing only the customer's last order. Marketing targets by city rather than by need. Leadership cannot answer a question as basic as "how many pharma manufacturers in Hyderabad buy from us?" without a week of spreadsheet work.

The problem statement

Design and build a customer knowledge graph: one connected model that resolves every signal about a customer, internal and external, into entities and relationships, and exposes it to a sales team through plain queries and ready-made briefs.

The company already holds four kinds of data for every customer:

- Order history: SKUs, HSN codes, quantities, values, invoice GSTIN, and billing and shipping addresses with PIN codes.
- Web behaviour: pages, categories, searches, carts, and time on site.
- Account details: company name, email domain, and phone.
- Session metadata: IP address, device, user agent, and mobile or desktop.

With effort, more is publicly available: GSTIN registration details (legal name, state, registration type), MCA company master data (incorporation date, directors, registered office), Udyam MSME registrations, import and export records, government tenders and awards, patents, funding and hiring news, company websites and LinkedIn pages.

The graph must turn all of this into answers to questions a human could not answer from the raw data. Six problems sit inside that:

1. Entity resolution: decide that "ABC Automation Pvt Ltd", "abc automation", a GSTIN, a Gmail account and a shipping PIN all refer to one company, and say how confident you are.
2. Enrichment from the internet: find and attach what open portals, scraped pages and third-party data services say about that company, and corroborate it across sources, because no single external source can be trusted on its own.
3. Graph modelling: a schema that connects accounts, people, locations, products, orders, sessions and external records, and survives new data types.

4. Inference: derive what is not stated anywhere, such as industry, project activity, buying stage, or the product a customer will need next.
5. Access for non-technical users: a salesperson on a phone, before a meeting, must get an answer without writing a query.
6. Freshness and trust: facts change, sources go down, and every claim must be traceable to where and when it came from.

How teams solve these is open. Graph databases, relational models, vector search, rules, classical ML and large language models are all acceptable, alone or combined. The company defines the opportunity. The team defines the solution.

The graph resolves internal and public fragments into one account, corroborates them across sources, scores them for confidence, and serves them back as the four things sales asks for.

Who the users are

The primary user is a salesperson with ten minutes before a meeting and a phone in hand. Everyone else is secondary.

User	What they need from the graph	A question they would ask
Field sales rep	A one-page brief before a visit	"What has this account bought in 12 months, what are they browsing now, and what projects are they running?"
Inside sales / telecalling	A daily list of who to call and why	"Which customers in this PIN code range have not ordered in 90 days?"
Marketing	Segments built on need, not geography	"Every drone-component buyer in Maharashtra with a GST registration, by industry"
Leadership	Territory and category planning	"Where are our top 100 automation customers concentrated, and who nearby looks like them?"

What a solution must achieve

The final product must power all four workflows below from one graph, on a dataset messy enough to prove the resolution works.

1. Account brief: given a company name, GSTIN, email domain or phone number, produce a brief a rep can read on a phone: who they are, where they are, what they buy, what they have been browsing, what the public record says they are working on, and what to pitch.
2. Territory and segment query: given a PIN code, city, district, industry or product category, return the matching customers ranked by value and recency, with enough context to plan a day of visits.

3. Lead discovery: given the existing customer base, surface companies in the public record that look like good customers but have never ordered, with the evidence for why.
4. Next-need signals: for a customer, infer what they are likely to need next from their own history, from similar customers, and from external signals such as a new import, tender or hire.

Across all four, the product must:

- Resolve identities across inconsistent names, personal email addresses, multiple shipping addresses and shared IPs, and expose a confidence score for every merge.
- Carry provenance: every fact shows which sources support it and when each was last seen.
- Degrade gracefully: a missing or unreachable source lowers confidence rather than breaking the answer.
- Be usable by a non-technical salesperson without training. A natural-language interface is welcome but not required. A well-designed form or search is acceptable.
- Update incrementally as new orders, sessions and external records arrive, without a full rebuild.

Data you can work with

The partner provides no internal company data. Teams design and generate a synthetic internal dataset that mirrors what the company holds, and it must be realistic in its mess: company names typed three ways, personal Gmail addresses on business accounts, several shipping PIN codes per customer, shared office IPs, and a GSTIN on some orders but not others.

Internal source to simulate	Fields
Orders	order id, date, SKUs, HSN codes, quantities, value, invoice GSTIN, billing and shipping address with PIN code
Web sessions	account or anonymous id, pages, categories, searches, cart events, timestamps
Accounts	company name as typed, contact name, email, phone, GST number if given
Session metadata	IP address, device type, user agent, mobile or desktop

Enrichment from the internet: allowed, and the hard part

Teams may use language models (LLMs and RLMs) and any enrichment source on the internet: open government portals, scraping of public web pages, and third-party data services that expose company information through APIs or exports. All are permitted within each source's terms of use, and data gathered for the challenge is used for the demonstration only.

Treat everything that comes from outside as a signal, never as a fact. Third-party services in India are incomplete and inconsistent. They publish no measure of their own accuracy, and there is no benchmark to check them against. The same holds for a scraped page or a model's extraction from one.

The central quality problem of this challenge is therefore corroboration rather than collection. A claim about a customer earns confidence when several independent sources agree, loses it when they disagree, and is marked uncertain when only one weak source makes it. The graph should record, for every fact, which sources said it, when, and how much weight it carries. How the team computes and shows that weight is its own design decision. That the weight exists and can be defended is required.

Open starting point	What it can tell you
GST portal (GSTIN search)	Legal name, trade name, state, registration type and status
MCA company master data	Incorporation date, registered office, directors, company status
Udyam registration	MSME classification and activity
Import and export records (customs and DGFT data, directly or through services that index it)	What a company imports, from where, how often
Tender portals (GeM, CPPP, state portals)	Government projects bid for and awarded
Patent office filings	Technology areas a company is developing
Company websites, LinkedIn pages, job posts, news	Products, projects, hiring, funding
India Post PIN directory, OpenStreetMap	PIN code to locality, city and district
CBIC HSN master	Product classification behind every order line
Third-party company data APIs	Aggregated profiles, usable as one signal among several

These are starting points, not requirements. A team that finds a better source and shows why it can be trusted is doing exactly what the challenge asks.

Scope, constraints and compliance

The challenge is the graph and the four workflows on top of it. It is not a CRM, not a storefront recommender, and not a scraper of individuals.

In scope	Out of scope
Resolving companies, contacts, locations and products into one graph	Replacing the company's CRM or order system
Enriching accounts from open and third-party sources, with corroboration and provenance	Product recommendations on the public website

In scope	Out of scope
Inferring industry, project activity and likely next need, with evidence	Building a general-purpose company database for India
Query and brief interfaces for sales, marketing and leadership	Profiling individuals from personal social media accounts

Personal data. Contact names, email addresses, phone numbers, IP addresses and device identifiers are personal data under India's Digital Personal Data Protection Act, 2023. Collect only what the workflow needs, keep it for a stated purpose, and show that a person's data can be found and deleted on request. Business identifiers such as a GSTIN, a registered office or a company's import record are not personal data, but the design should still show restraint about how they are combined with data about individuals.

Deployability. The partner will run this on its own infrastructure. The architecture must be deployable on private servers or a private cloud, with open-source components preferred. Any hosted service, including a hosted language model, must be optional and swappable. The product cannot depend on sending customer data to a third party to function.

Freshness. Sources change and go down. The design must separate what was observed from when it was observed, and must not present a two-year-old record as current fact.

Demonstration and evaluation

Each stage of CODERED '26 asks for a little more of the product, and the on-campus sprint asks for all of it.

Stage	What we want to see
Online round 1: PPT and video	The graph model, the sources chosen and why, the corroboration approach, and one workflow working end to end on the team's synthetic data
Online round 2: pitch	A live query: give the panel a company name or a PIN code and show the answer, with its sources and confidence
On-campus sprint: build and evaluate	All four workflows on a dataset with deliberate conflicts, plus a clear account of what the system got wrong and how it knows

Evaluators will weigh, in roughly this order:

1. Resolution quality: how well fragments are merged, measured on a labelled messy set the team builds, with the false-merge rate reported alongside the match rate.
2. Corroboration and provenance: whether every external fact carries its sources and a defensible confidence, and whether disagreement between sources is visible rather than hidden.
3. Usefulness to a salesperson: whether the brief and the query results change what a rep would do in the meeting.
4. Query experience: speed, clarity and how little the user has to know to ask.

5. Data handling: compliance with the personal-data and deployability constraints above.
6. Path to production: what it would take to connect real systems and keep the graph current.

What would impress:

- A measured match rate and false-merge rate on a hard labelled set.
- An explanation attached to every inferred fact.
- A brief that fits on one phone screen.
- Correct handling of Indian naming quirks such as “Pvt Ltd” versus “Private Limited”, transliterated names and abbreviations.
- PIN-to-district reasoning that survives a wrong digit.
- A system that says “unknown” when the evidence is thin instead of guessing.

What comes next

The Challenge Partner will consider standout team members for internships.

In line with CODERED’s student ownership principle, participation does not transfer ownership or commercial rights over the team’s work to the partner. Any licensing, collaboration or commercial engagement after the event is a separate discussion and agreement between the partner and the team.

Questions about this brief during the challenge go to E-Cell BMSITM at ecell@bmsit.in.